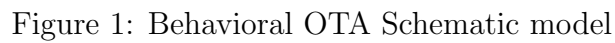


Negative Feedback

1.1 1. Schematic setup & Config View



1.2 2. Closed Loop Gain vs Frequency

AC Simulation Magnitude Data Table

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CIN=4p						
1	Lab08:OTA_behav_tb:1	y _{max} (dB20(VF("/VOUT")))	5.752			
1	Lab08:OTA_behav_tb:1	y _{max} (mag(VF("/VOUT")))	1.939			
1	Lab08:OTA_behav_tb:1	bandwidth(VF("/VOUT") 3 "lo...	2.519M			
1	Lab08:OTA_behav_tb:1	gainBwProd(VF("/VOUT"))	4.897M			
Parameters: CIN=12p						
2	Lab08:OTA_behav_tb:1	y _{max} (dB20(VF("/VOUT")))	11.51			
2	Lab08:OTA_behav_tb:1	y _{max} (mag(VF("/VOUT")))	3.764			
2	Lab08:OTA_behav_tb:1	bandwidth(VF("/VOUT") 3 "lo...	1.3M			
2	Lab08:OTA_behav_tb:1	gainBwProd(VF("/VOUT"))	4.905M			

Figure 3: AC Simulation parameters for $C_{IN} = 4\text{pF}$ and $C_{IN} = 12\text{pF}$.

Closed Loop Gain Comparison

Parameter	$C_{IN} = 4\text{pF}$		$C_{IN} = 12\text{pF}$	
	Hand Analysis	Simulation	Hand Analysis	Simulation
DC Gain (V/V)	2	1.939	4	3.764
DC Gain (dB)	6.02 dB	5.752 dB	12.04 dB	11.51 dB
Bandwidth	-	2.519 MHz	-	1.300 MHz
GBW	-	4.897 MHz	-	4.905 MHz

$$\beta = \frac{C_F}{C_F + C_{IN}}$$

Q: Comment on the difference between the results for the two values of C_{IN} :

Increasing C_{IN} from 4pF to 12pF decreases the feedback factor (β). Since the closed-loop gain is inversely proportional to β , the DC gain increases from ideally 2 to 4. Because both configs have the same open loop bandwidth and

$$BW_{CL} = BW_{OL}(1 + \beta A_{OL})$$

It is expected that $C_{IN} = 4\text{pF}$ gives higher closed loop bandwidth.

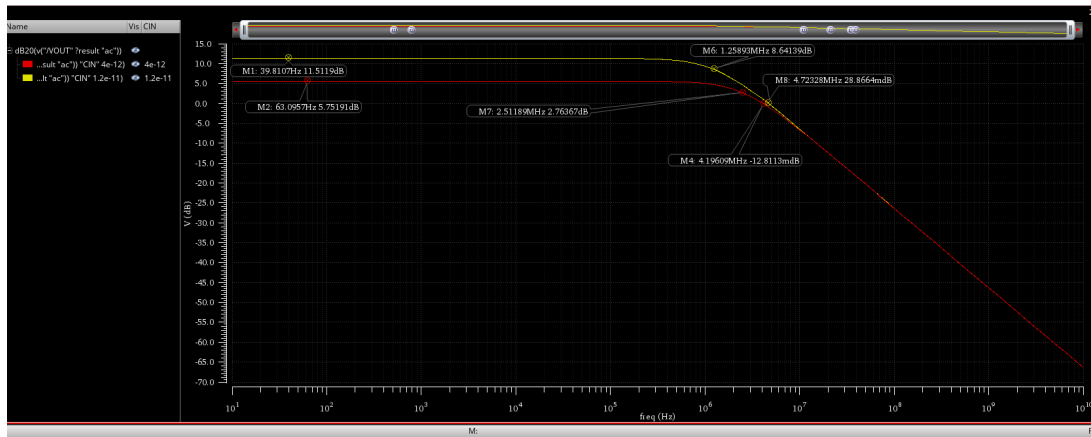


Figure 4: Closed-loop gain vs frequency

1.3 3. Loop Gain vs Frequency (STB Analysis)

STB Simulation Magnitude Plot (for $C_{IN} = 4pF$ and $12pF$)

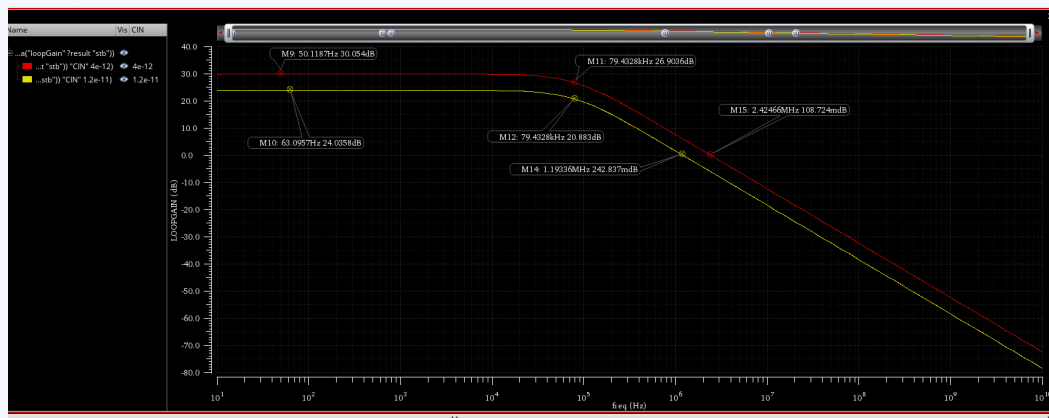


Figure 5: Loop Gain Magnitude versus Frequency comparing both feedback configurations.

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: $C_{IN}=4p$						
1	Lab08:OTA_behav_tb:1	unityGainFreq(mag(getData(...	2.459M			
1	Lab08:OTA_behav_tb:1	ymax(dB20(getData("loopGa...	30.06			
1	Lab08:OTA_behav_tb:1	bandwidth(getData("loopGai...	78.04k			
Parameters: $C_{IN}=12p$						
2	Lab08:OTA_behav_tb:1	unityGainFreq(mag(getData(...	1.228M			
2	Lab08:OTA_behav_tb:1	ymax(dB20(getData("loopGa...	24.04			
2	Lab08:OTA_behav_tb:1	bandwidth(getData("loopGai...	77.13k			

Figure 6: STB parameters

Loop Gain Comparison

Parameter	$C_{IN} = 4\text{pF}$		$C_{IN} = 12\text{pF}$	
	Hand Analysis	Simulation	Hand Analysis	Simulation
DC Loop Gain (dB)	30.05 dB	30.06 dB	24.036 dB	24.04 dB
Dominant Pole	-	78.04 kHz	-	77.13 kHz
Unity Gain Freq	-	2.459 MHz	-	1.228 MHz

Q: Comment on the differences between the results for the two values of C_{IN} :

The loop gain is defined as $A_{OL}\beta$. When C_{IN} increases, the feedback factor β drops. As a result, the DC loop gain decreases, which shifts the entire magnitude plot downwards, causing the loop gain to cross 0 dB at a lower Unity Gain Frequency (UGF) but not affecting the poles.

1.4 4. Gain Desensitization

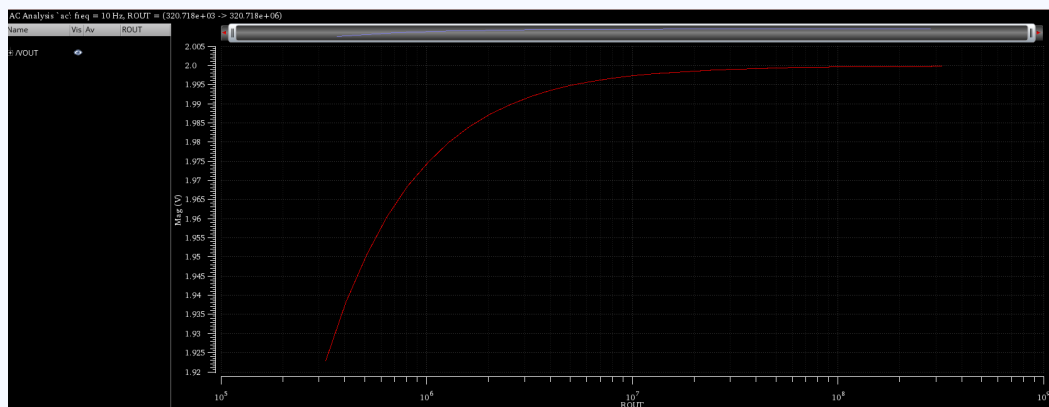
DC Closed Loop Gain (Magnitude) vs Open Loop Gain (A_v)

Figure 7: Sweep of output magnitude (Closed Loop Gain) across sweeping values of A_v at 10Hz.

Gain Desensitization Analysis

Q: Calculate the percent change in closed loop gain (magnitude). Note that open loop gain (A_v) changes by three orders of magnitude. Comment.

- $A_{v,min} = 50$, Closed-Loop Gain = 1.922 V/V
- $A_{v,max} = 50000$, Closed-Loop Gain = 2.000 V/V

- **Percent Change** = 4.06%

Even though the open-loop gain varies drastically (by a factor of 1000), the closed-loop gain remains almost entirely stable at approximately 2 V/V. This demonstrates the primary advantage of negative feedback: it desensitizes the circuit's overall performance from variations in the active amplifier's internal parameters.

2. Part 2: Feedback with Real 5T OTA

Real Transistor-Level OTA Implementation

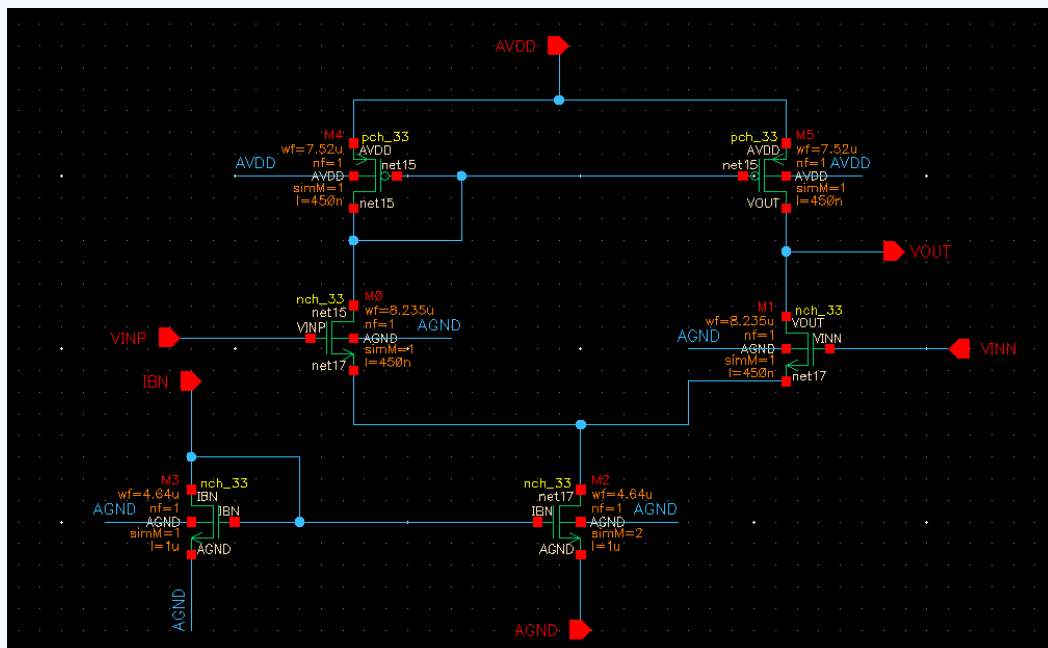


Figure 8: Schematic implementation of the real 5T Operational Transconductance Amplifier.

2.1 1. Closed Loop Gain vs Frequency

AC Simulation Magnitude Plot (Real OTA)

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CIN=4p						
1	Lab08:OTA_behav_tb:1	y _{max} (dB20(VF("/VOUT")))	5.898			
1	Lab08:OTA_behav_tb:1	y _{max} (mag(VF("/VOUT")))	1.972			
1	Lab08:OTA_behav_tb:1	bandwidth(VF("/VOUT")) 3 "lo...	1.759M			
1	Lab08:OTA_behav_tb:1	gainBwProd(VF("/VOUT"))	3.476M			
Parameters: CIN=12p						
2	Lab08:OTA_behav_tb:1	y _{max} (dB20(VF("/VOUT")))	11.66			
2	Lab08:OTA_behav_tb:1	y _{max} (mag(VF("/VOUT")))	3.83			
2	Lab08:OTA_behav_tb:1	bandwidth(VF("/VOUT")) 3 "lo...	789.3k			
2	Lab08:OTA_behav_tb:1	gainBwProd(VF("/VOUT"))	3.03M			

Figure 9: Closed Loop AC results

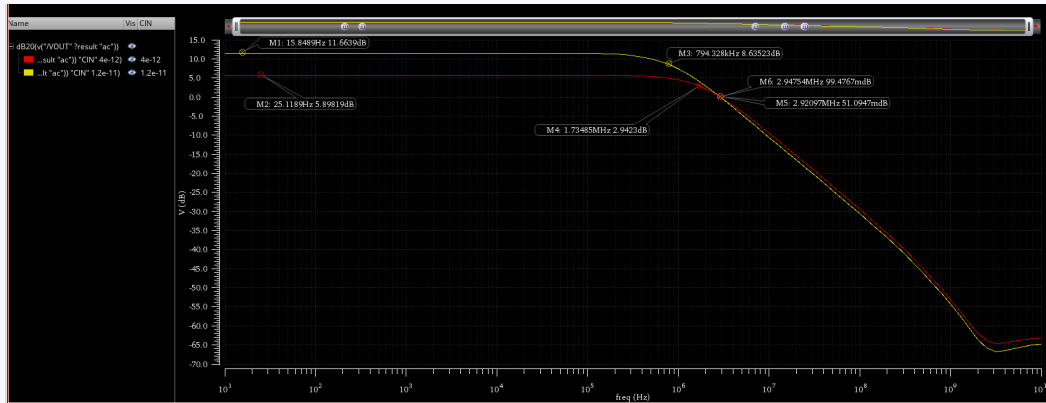


Figure 10: Closed Loop AC response

Closed Loop Gain Comparison (Behavioral vs. Real)

Parameter ($C_{IN} = 4pF$)	Part 1 (Behavioral)	Part 2 (Real 5T OTA)
DC Gain (V/V)	1.939 V/V	1.972 V/V
Bandwidth	2.519 MHz	1.759 MHz
GBW	4.897 MHz	3.476 MHz

Q: You will notice that the bandwidth, and consequently the GBW are much smaller than Part 1. Why? Comment.

Unlike the ideal behavioral model which uses a perfect buffer to isolate the feedback network, the real 5T OTA has a finite output resistance and physical parasitic capacitances. The feedback network (C_F and C_{IN}) heavily loads the output node of the OTA, dropping the dominant pole to a lower frequency and drastically reducing the operational bandwidth.

2.2 2. Loop Gain vs Frequency (STB Analysis)

STB Simulation Magnitude Data Table (Real OTA)

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CIN=4p						
1	Lab08:OTA_behav_tb:1	unityGainFreq(mag(getData(...	1.72M			
1	Lab08:OTA_behav_tb:1	ymax(dB20(getData("loopGa...	30.24			
1	Lab08:OTA_behav_tb:1	bandwidth(getData("loopGai...	53.23k			
Parameters: CIN=12p						
2	Lab08:OTA_behav_tb:1	unityGainFreq(mag(getData(...	751k			
2	Lab08:OTA_behav_tb:1	ymax(dB20(getData("loopGa...	24.28			
2	Lab08:OTA_behav_tb:1	bandwidth(getData("loopGai...	45.73k			

Figure 11: Calculator extractions for Unity Gain Frequency and DC loop gain

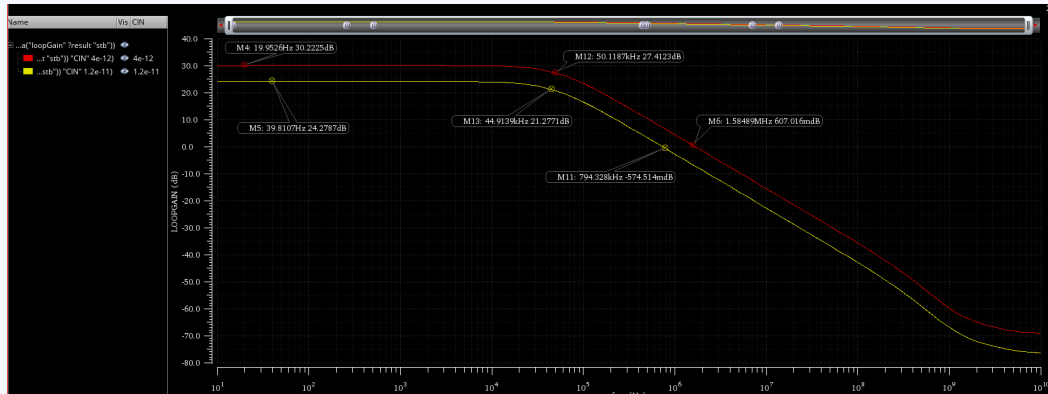


Figure 12: Loop Gain vs frequency

Loop Gain Comparison (Behavioral vs. Real)

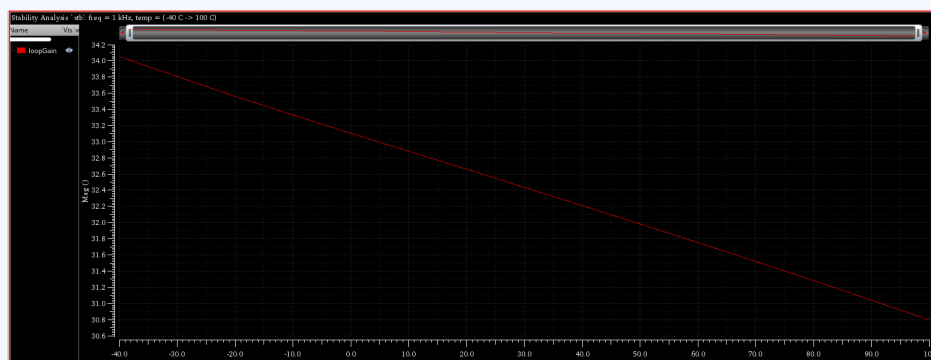
Parameter ($C_{IN} = 4pF$)	Part 1 (Behavioral)	Part 2 (Real 5T OTA)
DC Loop Gain	30.06 dB	30.24 dB
Unity Gain Frequency	2.459 MHz	1.720 MHz
Dominant Pole	78.04 kHz	53.23 kHz

Q: You will notice that the unity gain frequency is much smaller than Part 1. Why? Comment.

In the real 5T OTA, the unity-gain frequency of the loop gain is severely degraded because the capacitive voltage divider acts as an extra load on the output. Furthermore, internal parasitic poles introduce additional phase shift and magnitude roll-off that the ideal single-pole behavioral model simply does not possess.

2.3 3. Gain Desensitization over Temperature

DC Loop Gain vs Temperature

Figure 13: Tracking the decay of the DC loop gain over a temperature range of -40°C to 100°C .

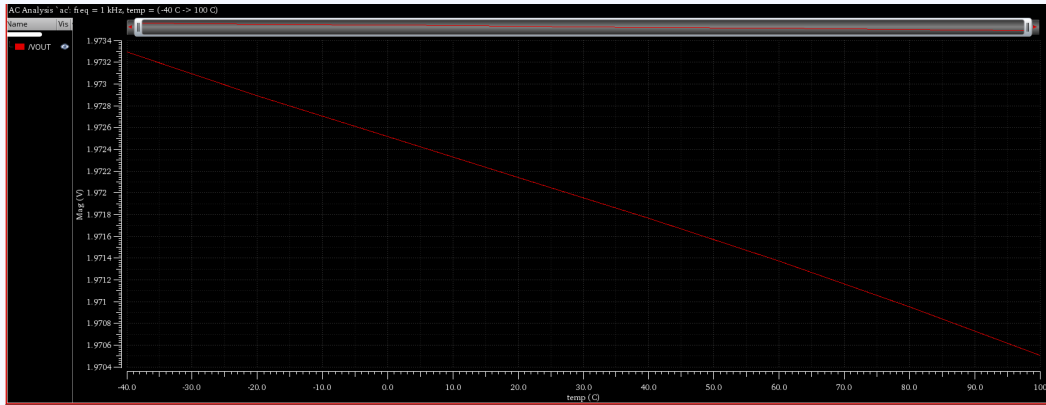


Figure 14: Closed Loop Gain vs temperature

Gain Desensitization Analysis

Q: Compare the percent change in the DC loop gain (from STB) and the DC closed loop gain (from AC) across temperature extremes.

- **Loop Gain:** $LG_{-40^{\circ}C} = 34.05$, $LG_{100^{\circ}C} = 30.80 \rightarrow$ **Change:** -3.25 ($\approx -9.545\%$ linear drop)
- **Closed Loop Gain:** $A_{CL,-40^{\circ}C} \approx 1.9733 \text{ V/V}$, $A_{CL,100^{\circ}C} \approx 1.9705 \text{ V/V} \rightarrow$ **Change:** 0.1419%

While temperature variations drastically affect the internal transistor parameters (like threshold voltage and mobility) leading to massive shifts in the open-loop gain, the negative feedback network enforces a nearly constant closed-loop response.

2.4 4. Transient Analysis

Transient Plots: $V_P - V_N$ for 1kHz

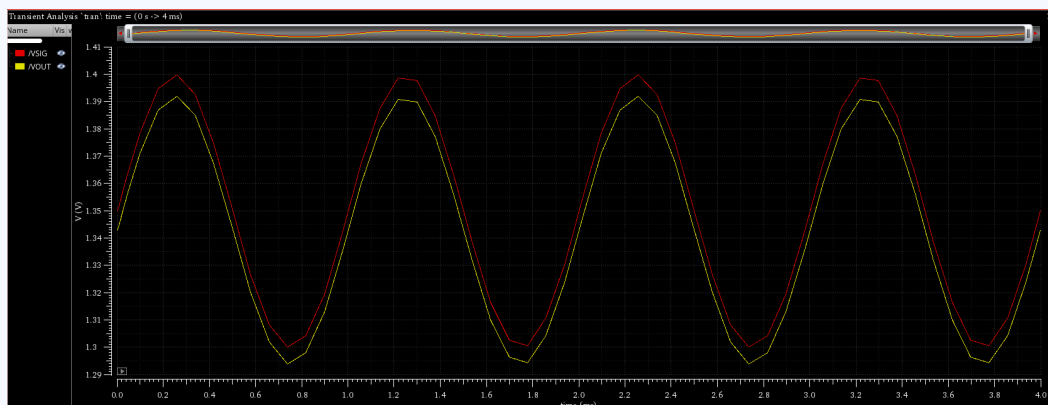


Figure 15: Vout vs Vs at FIN=1kHz

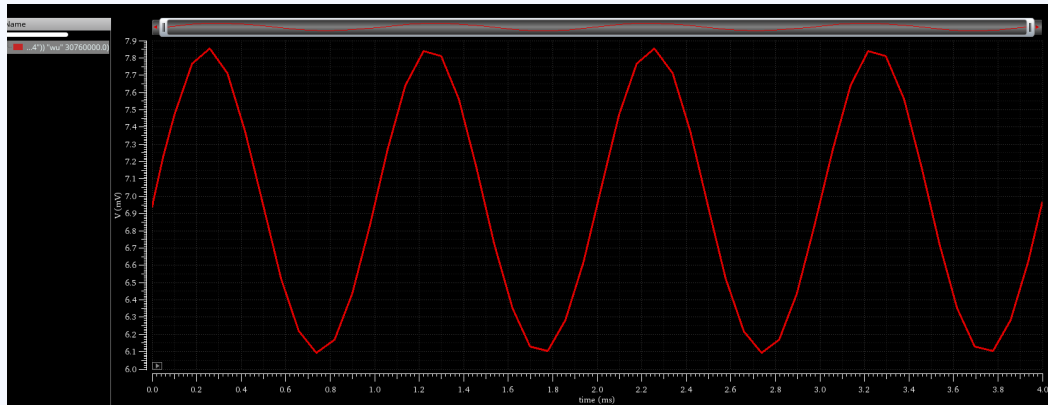


Figure 16: Input Differential vs time

Transient Plots: $V_P - V_N$ for 1.759MHz

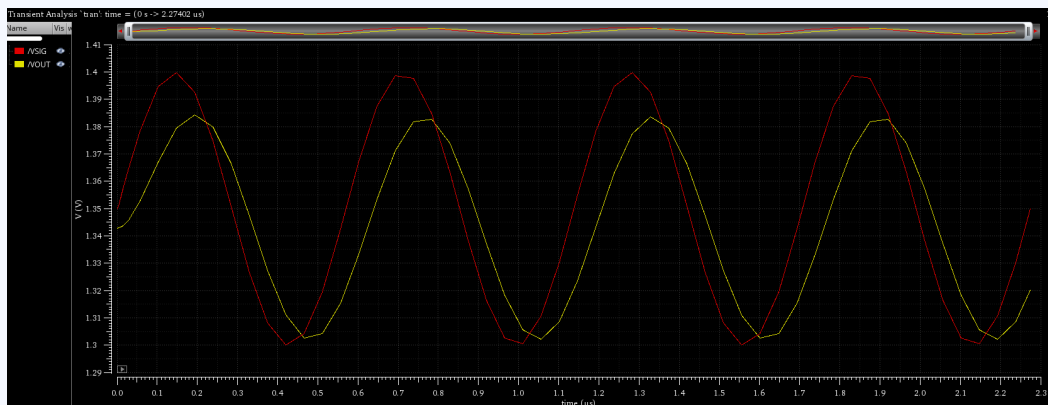


Figure 17: Vout vs Vsig at FIN=1.759MHz

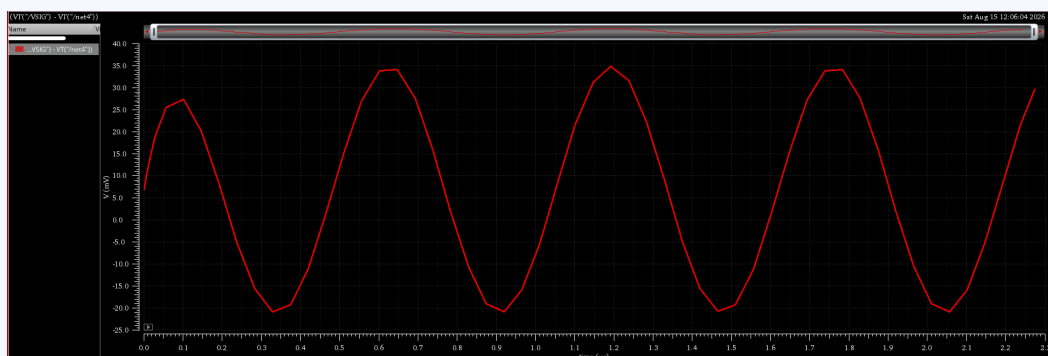


Figure 18: Input Differential vs time at FIN=1.759MHz

Transient Analysis & Open Loop Relation

At $F_{IN} = 1\text{kHz}$:

- $V_{in,pp} = 100 \text{ mV}$
- $V_{out,pp} \approx 193.6 \text{ mV}$
- $(V_P - V_N)_{pp} = 1.75 \text{ mV}$

Relation: The output relates to the differential input via the open-loop gain: $V_{out} \approx A_{OL} \cdot (V_P - V_N)$. Since loop gain isn't degraded at 1kHz, $(V_P - V_N)$ is extremely small.

At $F_{IN} = \text{Closed Loop Bandwidth (1.759MHz)}$:

- $V_{in,pp} = 100 \text{ mV}$
- $V_{out,pp} = 80 \text{ mV}$
- $(V_P - V_N)_{pp} = 55.7822 \text{ mV}$

Relation: At the closed-loop bandwidth, the output signal is attenuated by 3 dB relative to the ideal DC closed-loop gain, and a significant phase shift is introduced between V_{in} and V_{out} . Furthermore, because F_{IN} is beyond the unity gain frequency of the loop gain, the negative feedback is degraded. The virtual short breaks down, meaning the error signal $(V_P - V_N)$ becomes large (reaching 55.78 mV) as it must increase to sustain the output, unlike the 1 kHz case where $(V_P - V_N)$ was nearly zero.